

Hypothesis testing

Exploring Data and setting up for an analysis

Joseph Mhango

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Objectives

- Question formulation
- Summarize: Weighing the Pig
- Variables and graphing
- “Analysis” versus “EDA”
- Statistical Analysis Plan: the concept

Question formulation and hypothesis testing

- “population of interest”
- samples and sampling
- test statistics
- null hypothesis
- Let's talk about the P-value

Hypotheses: Null vs Alternative

- **Null Hypothesis (H_0):**
The assumption that there is **no effect** or **no difference**.
Example: “There is no difference in test scores between two groups.”
- **Alternative Hypothesis (H_1 or H_a):**
The assumption that there **is an effect** or **a difference**.
Example: “There is a difference in test scores between two groups.”

P-value: Definition and Interpretation

- **P-value:**

The probability of obtaining results at least as extreme as the ones observed, under the assumption that the null hypothesis is true.

- **Interpretation:**

A **small p-value** (typically < 0.05) suggests that the observed data is unlikely under the null hypothesis, leading to rejection of H_0 .

A **large p-value** suggests that the data is consistent with the null hypothesis.

Question formulation and hypothesis testing

Benefits of NHST

- Familiar and acceptable to researchers
- Typically robust to assumptions
- Strong framework for evidence
- The basic idea is simple

Question formulation and hypothesis testing

Criticism of NHST

- Often interpreted under error
- Validation of analysis often neglected
- Education often deficient
- Practitioners ignorant of subtleties

Summarize: Weighing the Pig

Chick weight dataset

The hypothesis voices "how you think the world works" or what you predict to be true"

coding

Variables and graphing



Variables and graphing

- Must convey relevant information
- Consistent in aesthetics
- Self-contained
- Reflect hypothesis (unless descriptive)
- Appropriate to data

Variables and graphing

Concept of “layering” in building graphs

coding

“Analysis” versus “EDA”

EDA:

- Informal, haphazard
- Gain data understanding
- Test assumptions
- Usually not for “others”
- Usually occurs before analysis

“Analysis” versus “EDA”

Analysis:

- Designed to fit hypothesis
- For presentation to others
- Creation of EVIDENCE to support CLAIMS
- Reproducible

Statistical Analysis Plan: the concept

Prior to data collection

- formally state hypothesis
- State specific statistical model(s)
- Specify data and data collection
- State and justify sample size

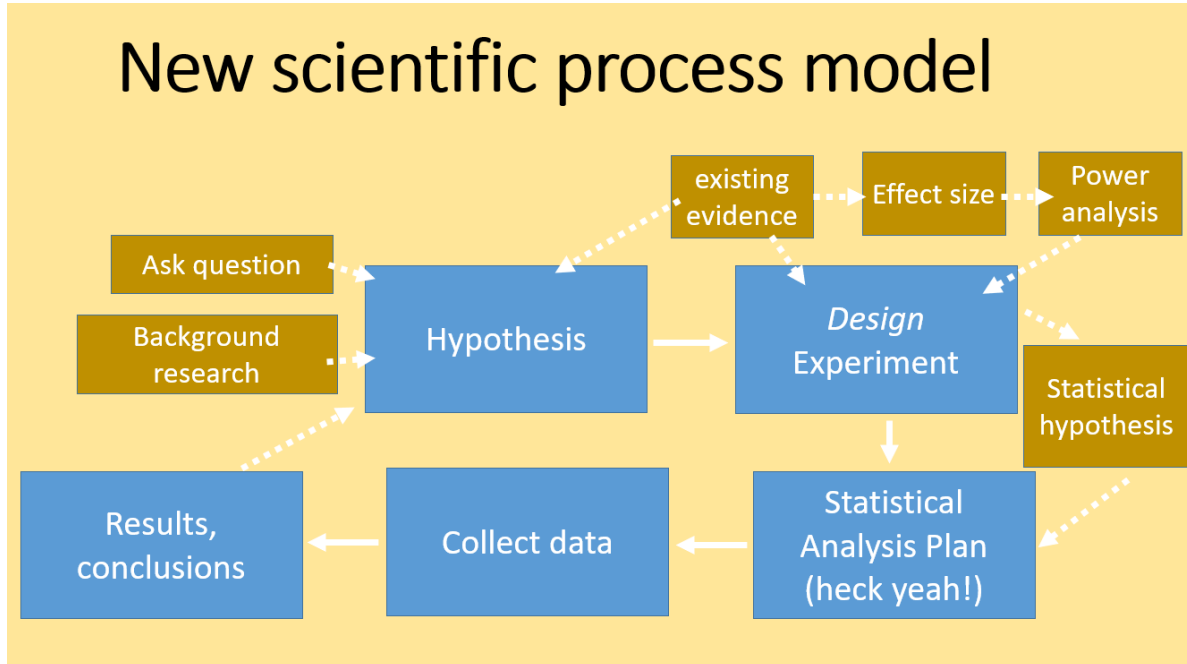
Taught to children

(Ye Olde) Scientific Process model

The scientific process model we teach to children



Best practice



Practice Exercises

